

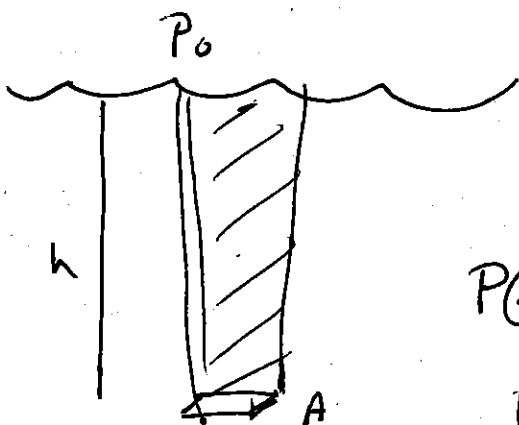
14 Fluids

1/10

$$P = F/A \quad \text{N/m}^2 \quad 1 \text{ N/m}^2 = 1 \text{ Pa}$$

$$P_0 = 1.013 \cdot 10^5 \text{ Pa} \quad \text{air pressure at sea level}$$

discuss gauge vs absolute pressure.



$$P(h) = P_0 + \frac{\rho g h A}{A}$$

$$\boxed{P(h) = P_0 + \rho g h}$$

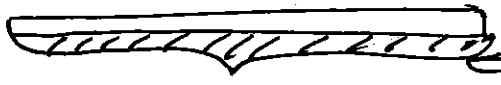
$$\rho_w = 1000 \text{ kg/m}^3 \quad \rho_{sw} = 1030 \text{ kg/m}^3 \quad \rho_{ice} = 917 \text{ kg/m}^3$$

$$\begin{aligned} P(10 \text{ m}) &= P_0 + \rho g h = 1.013 \cdot 10^5 + 1000 \cdot 9.8 \cdot 10 \\ &= 1.013 \cdot 10^5 + 9.8 \cdot 10^4 \text{ N/m}^2 \\ &= 1.993 \cdot 10^5 \text{ N/m}^2 \sim 2 P_0 \end{aligned}$$

2/10



displaced water



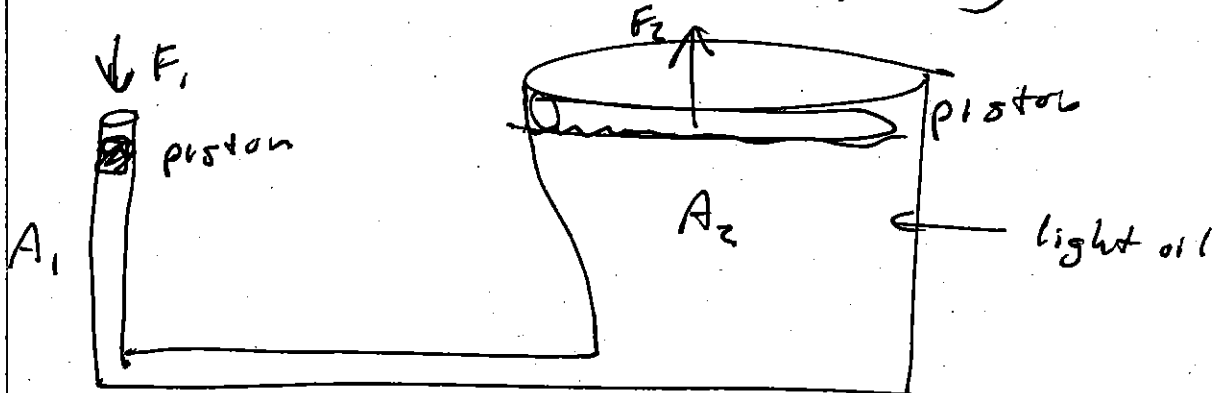
more water displaced

Archimedes' Principle

Buoyant force = weight of water displaced

DeVinci Code

~ 250BC Gold crown Sicily (Syracuse) Hero



$$P = \frac{F_1}{A_1} = \frac{F_2}{A_2}$$

$$A_1 = 10^{-4} \text{ m}^2$$

$$F_2 = F_1 \frac{A_2}{A_1}$$

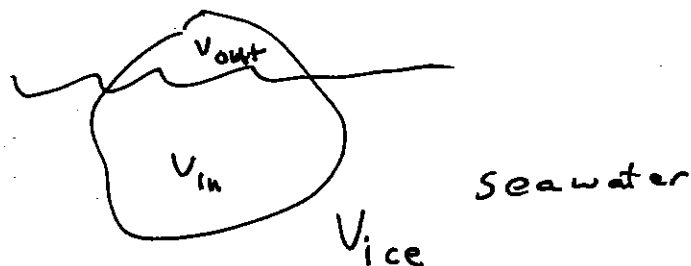
$$A_2 = 1 \text{ m}^2$$

$$F_2 = 10^4 F_1 \quad 10^4 \text{ magnification!}$$

3/10

~~iceberg~~

iceberg



$$V_{ice} = V_{in} + V_{out}$$

$$\uparrow B = V_{in} \rho_{sw} g$$

$$\downarrow W = mg = V_{ice} \rho_{ice} g$$

$$V_{in} \rho_{sw} g = V_{ice} \rho_{ice} g$$

$$\frac{V_{in}}{V_{ice}} = \frac{\rho_{ice}}{\rho_{sw}} = \frac{917}{1030} = 89.03\%$$

→ 11% out

in fresh water

8% is above water

4/10

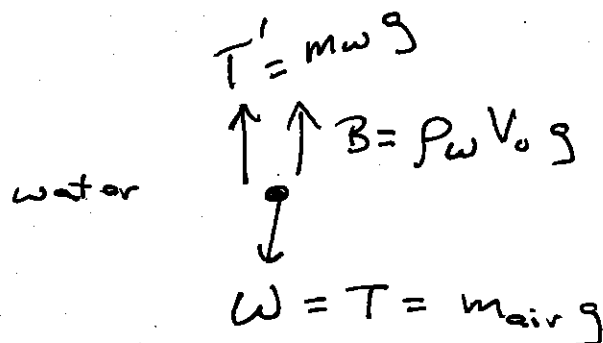
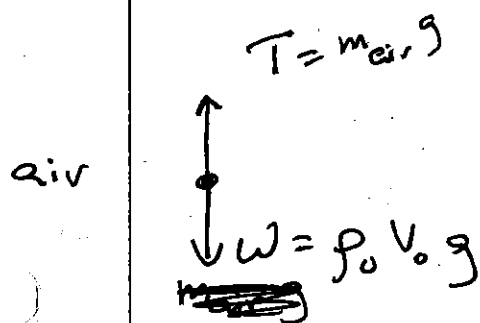
Geologist assaying a rock sample.

Weights it on a spring scale in air

$$m_{\text{air}} = 2 \text{ kg}$$

and weights it in water

$$m_w = 1 \text{ kg}$$



$$m_{\text{air}} = \rho_0 V_0$$

2 unknowns!

$$\rho_0 = \frac{m_{\text{air}}}{V_0} = \frac{2 \text{ kg}}{10^{-3} \text{ m}^3}$$

$$\rho_0 = 2 \cdot 10^3 \text{ kg/m}^3$$

$$m_w + \rho_w V_0 = m_{\text{air}}$$

1 unknown V_0

$$V_0 = \frac{m_{\text{air}} - m_w}{\rho_w}$$

$$V_0 = \frac{2 \text{ kg} - 1 \text{ kg}}{1000 \text{ kg/m}^3}$$

$$= \frac{1}{1000} \text{ m}^3 = 10^{-3} \text{ m}^3$$

EFFI Al has $\rho = 2.7 \cdot 10^3 \text{ kg/m}^3$

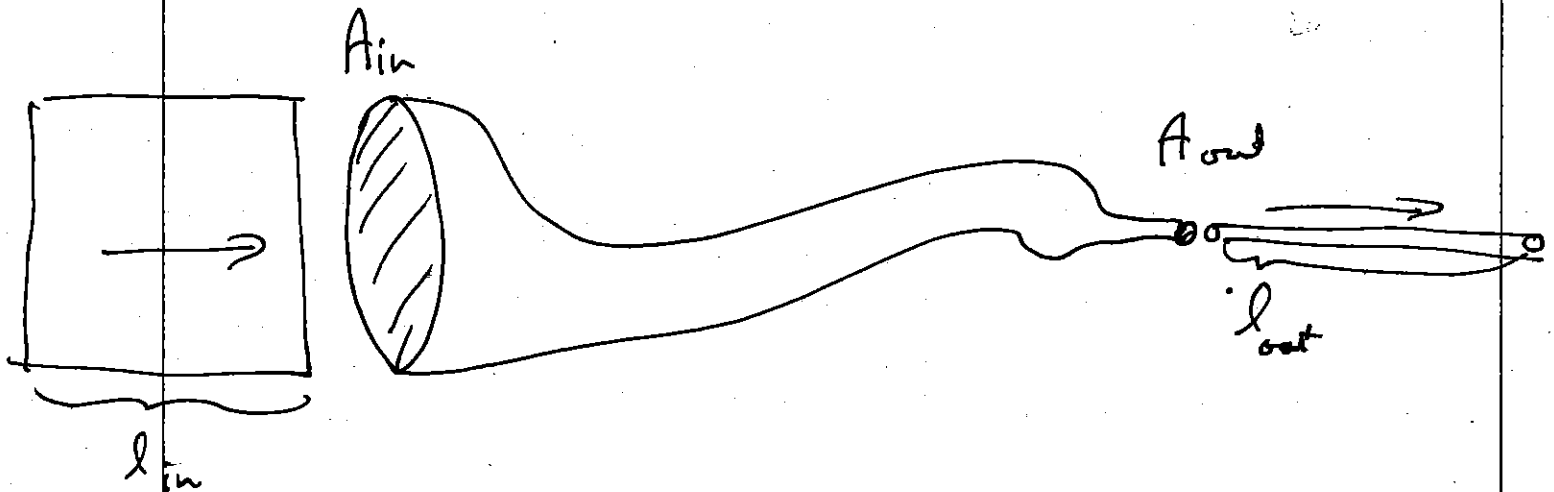
Dynamic Fluids

5/10

Mass in equals mass out

Continuity eq.

assume ρ is constant



$$\rho A_{in} l_{in} = \rho A_{out} l_{out}$$

Al is volume

$$l/t = v$$

$$\rho A_{in} v_{in} = \rho A_{out} v_{out}$$

$$A_{in} v_{in} = A_{out} v_{out}$$

$$v_{out} = \frac{A_{in}}{A_{out}} v_{in}$$

if A_{in} is circle of radius 1cm. $A_{in} = \pi 0.01^2$

if A_{out} is 0.5mm radius $A_{out} = \pi 0.0005^2$

$v_{in} = \frac{1 \text{ m/s}}{10 \text{ s}} = \frac{0.1 \text{ m}}{10 \text{ s}} = 0.01 \text{ m/s}$

$v_{out} = \frac{0.01 \text{ m/s}}{\frac{\pi 0.01^2}{\pi 0.0005^2}} = \frac{0.01 \text{ m/s}}{400} = 2.5 \times 10^{-5} \text{ m/s}$

Demos

Bernoulli Effect

Cons of Energy (again)

remember Energy is force · distance work
done on a fluid

$$E = Fd = \frac{Fd A}{A} = PV$$

$$\frac{F}{A} = P \quad dA = V$$

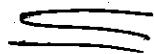
$$KE = \frac{1}{2} \rho V^2 \quad \frac{1}{2} m v^2 \quad m = \rho V$$

$$\frac{1}{2} \rho V v^2$$

$$PE = \rho g y \quad \cancel{m} m g h = \rho V g h$$

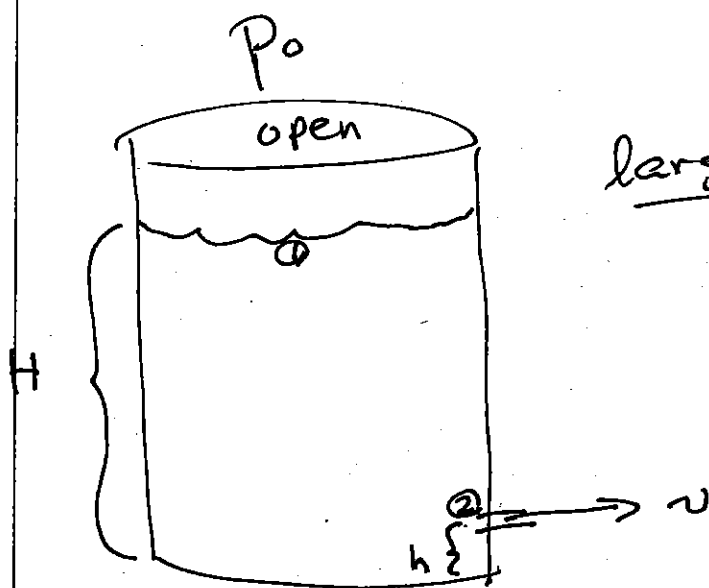
$$P_1 + \rho g y_1 + \frac{1}{2} \rho v_1^2 = P_2 + \rho g y_2 + \frac{1}{2} \rho v_2^2$$

Energy density!



Consider open water tower

7/10



large open water tower.

$$P_1 + \rho g y_1 + \frac{1}{2} \rho v_1^2 = P_2 + \rho g y_2 + \frac{1}{2} \rho v_2^2$$

$$P_1 = P_0 \quad \text{open}$$

$$v_1 = 0 \quad \text{large}$$

$$y_1 = H = 30 \text{ m}$$

$$P_2 = P_0 \quad \text{open}$$

$$v_2 = ? \quad \text{variable sought}$$

$$y_2 = h = 1 \text{ m}$$

$$P_0 + \rho g H + \frac{1}{2} \rho 0^2 = P_0 + \rho g h + \frac{1}{2} \rho v^2$$

$$\rho g H = \rho g h + \frac{1}{2} \rho v^2$$

$$g H = g h + \frac{1}{2} v^2$$

$$v = \sqrt{2g(H-h)}$$

coyote off cliff again!

$$v = \sqrt{2 \cdot 9.8 \frac{\text{m}}{\text{s}^2} \cdot 29 \text{ m}} = 23.8 \text{ m/s}$$

fluid density ρ canceled out
all one fluid

8/10

P rises as you go deeper in tank

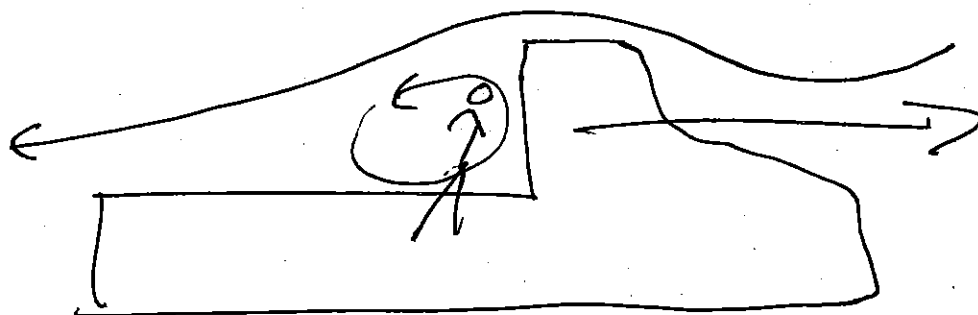
Air density is variable $1.21 \frac{\text{kg}}{\text{m}^3} - 1.29 \frac{\text{kg}}{\text{m}^3}$

depending on
altitude and temperature

Airplane wing

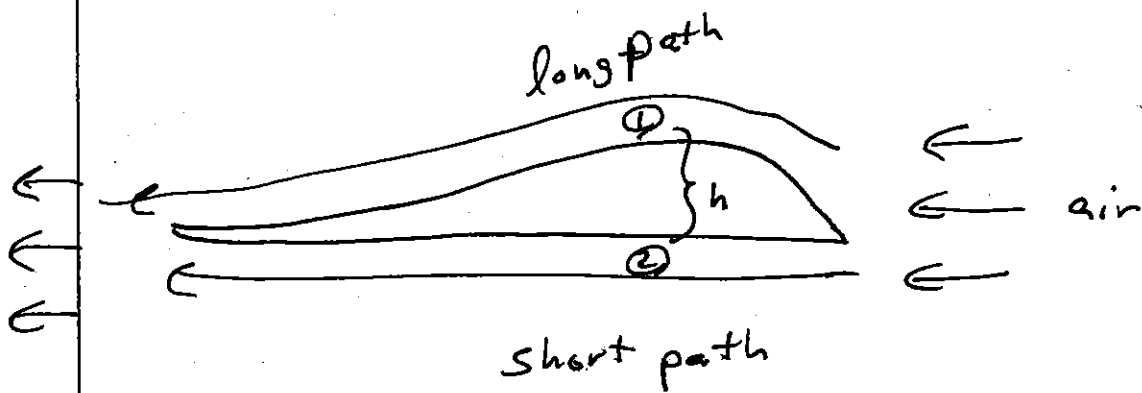
assumes laminar flow (no turbulence)

Pickup has turbulence



cavitation behind cab or gate
air gate or solid gate, or open
matters!

9/10



$$v_1 = 100 \text{ m/s} \quad v_2 = 80 \text{ m/s} \quad A = 100 \text{ m}^2$$

$$\rho = 1.21 \text{ kg/m}^3$$

$$P_1 + \rho g y_1 + \frac{1}{2} \rho v_1^2 = P_2 + \rho g y_2 + \frac{1}{2} \rho v_2^2$$

P is dynamic (it's open, but not P_0)!

$$P_1 + \frac{1}{2} \rho v_1^2 = P_2 + \frac{1}{2} \rho v_2^2$$

$$y_1 \sim y_2$$

thickness of wing is

$$\frac{1}{2} \rho (v_1^2 - v_2^2) = P_2 - P_1 > 0 \quad \text{not critical to flight.}$$

Lift = PA force A is area of wings

$$L = A (P_2 - P_1) = \frac{1}{2} \rho A (v_1^2 - v_2^2)$$

$$L = \frac{1}{2} \cdot 1.21 \frac{\text{kg}}{\text{m}^3} \cdot 100 \text{ m}^2 \cdot (100^2 - 80^2) \frac{\text{m}^2}{\text{s}^2}$$

$$L = 2.178 \cdot 10^5 \text{ N} \quad \text{lifts } m = 2.2 \cdot 10^4 \text{ kg}$$

$$\sim 50,000 \text{ lb}$$

Phoenix airport in summer closes
as ρ drops too low for lift off!

10/10

Smoking in car

inside ① | outside ②
cracked window

$$P_1 + \rho g y_1 + \frac{1}{2} \rho v_1^2 = P_2 + \rho g y_2 + \frac{1}{2} \rho v_2^2$$

$$y_1 = y_2$$

$$v_1 = 0$$

$$P_1 = P_2 + \frac{1}{2} \rho v_2^2$$

v_2 is car's speed

$$P_1 - P_2 = \frac{1}{2} \rho v_2^2$$

$$\sim 30 \text{ m/s}$$

$$\Delta P = \frac{1}{2} 1.21 \frac{\text{kg}}{\text{m}^3} \left(30 \frac{\text{m}}{\text{s}}\right)^2$$

$$= 544.5 \text{ Pa} \quad \text{air weakly flows out.}$$

"moving air has low pressure"